

Bike Sharing Demand Analysis: Summary Report

NextHikes IT Solutions · Project 1: Data Analysis with Excel

Executive summary

Bike rentals in this dataset are driven by commuters. Registered (subscribed) riders made 53,383 of the 58,304 rentals (91.6%), and demand peaks at 8 AM and 5 PM on weekdays.

Weather matters mainly when it rains: light rain or snow cuts hourly rentals by about 46%.

The analysis covers 1,000 hourly records from 1 January to 14 February 2011. It was built in Excel from three source datasets that were cleaned, merged and appended into one final table.

The main findings are:

- **Two commuter peaks.** The busiest hours are 5 PM (about 140 rentals per hour) and 8 AM (about 128). Between 1 AM and 5 AM fewer than 10 bikes an hour are rented.
- **Weekdays beat weekends.** Weekdays average 63.3 rentals per hour against 48.1 at weekends. Casual riders, however, more than double at weekends (8.2 vs 3.3 per hour).
- **Weekday commute hours carry 43% of all rentals.** These are 7 to 9 AM and 4 to 7 PM on working days.
- **Rain hurts, temperature helps a little.** Clear hours average 61.7 rentals, light rain/snow hours 33.2. Warm hours average 85.7 against 47.7 in cold hours, but the correlation of temperature with demand is weak (0.23).
- **Seven unusual hours** stood out in anomaly detection. They are real busy hours, so they were kept.

The recommendations are to keep bikes available at commuter stations before 8 AM and 5 PM, to use the night window for rebalancing and repairs, and to grow weekend leisure use with offers for casual riders.

Business problem and objectives

A bike-sharing operator needs to know when and why people rent bikes, so it can have bikes in the right place, plan staff and target marketing. The project asks how weather, time of day, day of week and holidays affect hourly rentals.

The objectives were to:

1. Clean and combine three raw datasets into one reliable table.

2. Use Excel text, lookup, conditional and statistical functions to describe demand.
3. Summarise demand by time, weather and rider type, and build an interactive dashboard.
4. Detect unusual hours and forecast the next 48 hours.
5. Turn the results into practical recommendations, with insights written as cell comments in the workbook.

Data overview

The three source files share the key column `instant`, a running number for each hour. Together they give 1,000 hourly records from 1 January to 14 February 2011, all in season 1.

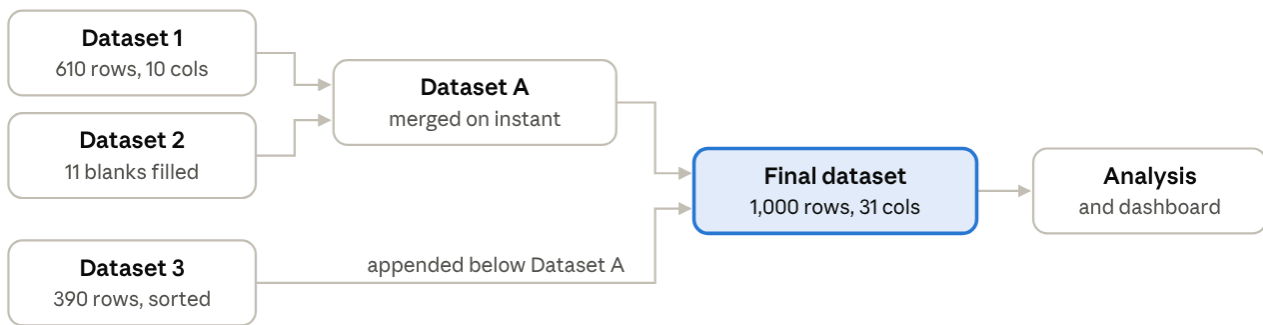
Dataset	Rows × columns	What it holds	Instant range
Dataset 1	610 × 10	Date, season, year, month, hour, holiday, weekday, weather code, temperature	1 to 610
Dataset 2	610 × 7 (after cleaning)	Feels-like temperature, humidity, wind speed, casual, registered and total rentals	1 to 610
Dataset 3	390 × 16	All 16 fields for the later hours	611 to 1000

Temperature, feels-like temperature, humidity and wind speed are normalised values between 0 and 1. The units are not given, so they were analysed as they are, without converting to °C or °F. The weather code means 1 Clear, 2 Cloudy/Mist, 3 Light Rain/Snow and 4 Heavy Rain.

Data preparation

Each dataset was checked on its own before combining, so the final table has 1,000 unique hours with no blanks and no duplicates.

Three raw files become one 1,000-row table



data preparation flow · 3 source files to 1 final table

1. **Dataset 1:** types checked (date, TRUE/FALSE, numbers). No blanks or duplicates.
2. **Dataset 2:** the column "Unnamed: 0" was a leftover row index, so it was removed. 11 blank feels-like temperature values were filled with the average of the hour before and the hour after; this is safe because it changes smoothly hour to hour. Each filled cell is highlighted and has a comment.
3. **Dataset 3:** sorted by `instant` (the file started at 620). No blanks or duplicates.
4. **Merge:** Dataset 1 and Dataset 2 were joined side by side on `instant` with INDEX/MATCH, which gives the same result as a Power Query Left Outer merge. All 610 keys matched, giving Dataset A (610 × 16).
5. **Append:** Dataset 3 was added under Dataset A to give the final table (1,000 × 16).
6. **Features:** 15 new columns were added with formulas, such as day type, weather label, time band, temperature band, peak-hour flag, casual share and demand level.

The Data_Quality sheet repeats these checks with formulas at every stage. It also confirms that total rentals equal casual plus registered on every row. Outliers were kept and flagged rather than deleted, and temperatures were not converted to other units.

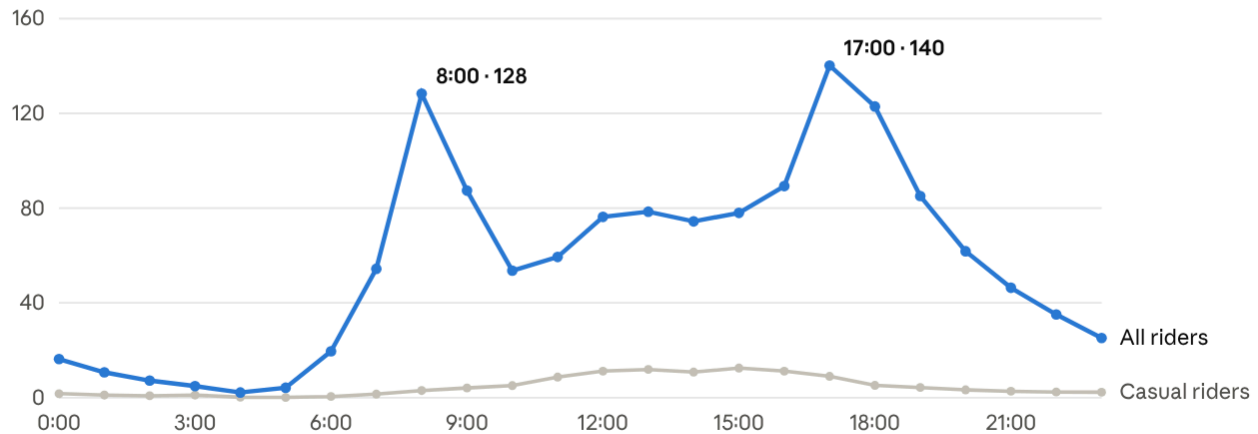
Analysis and findings

Overall the data has 58,304 rentals, an average of 58.3 per hour and a median of 49. Hour of day explains demand far better than weather.

Time of day

Rentals peak at 8 AM and 5 PM, the commute hours

Average rentals per hour, by hour of day, 1 Jan to 14 Feb 2011



Final_Dataset, 1,000 hourly records · Analysis sheet, table 1

Registered riders make both commute peaks. Casual riders are few and peak in the early afternoon (about 12 per hour between 12 and 4 PM). By time band, the evening (5 to 8 PM) averages 102.7 rentals per hour and the night (midnight to 5 AM) only 8.0.

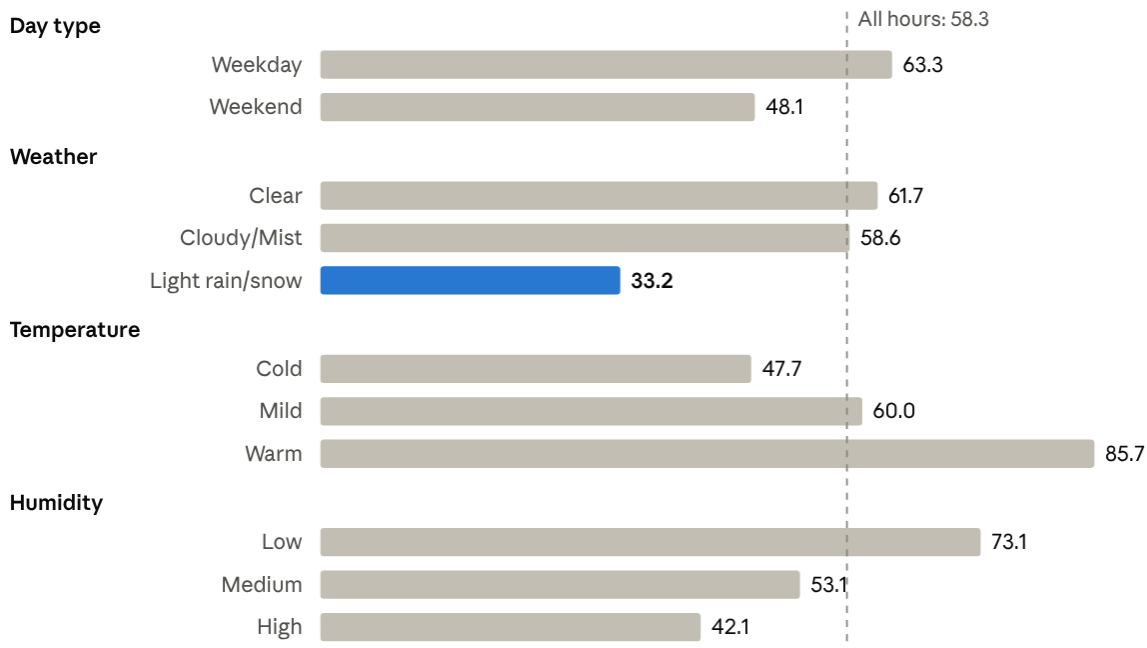
Day of week

Tuesday to Friday are the strongest days, at 61 to 68 rentals per hour; Saturday is the weakest at 46.7. At weekends registered use falls from 60.0 to 39.9 per hour, while casual use rises from 3.3 to 8.2. Weekends are leisure time, weekdays are commuting time.

Weather, temperature and humidity

Light rain cuts hourly rentals almost in half compared with clear weather

Average rentals per hour under each condition; dashed line = average of all hours



Final_Dataset, 1,000 hourly records · Analysis sheet, tables 2, 4, 6 and 7; heavy rain (1 hour) left out

Clear and cloudy hours are almost the same. Only rain makes a big difference. Warm hours (normalised temperature 0.30 or above) attract the most casual riders, 14.3 per hour against 2.3 in cold hours. Heavy rain appears in a single hour, which is too little to draw a conclusion.

Relationships between variables

The correlation matrix in the Stats sheet shows weak links between weather and demand: temperature 0.23, humidity -0.25, wind speed 0.10. Temperature and feels-like temperature are almost the same signal (0.93), so only one is needed. Registered rentals track total rentals almost perfectly (0.99). Correlation shows association, not cause.

Where rentals concentrate

Of the 58,304 rentals, 41,435 (71%) were on working days and 25,033 (43%) in weekday commute hours (7 to 9 AM and 4 to 7 PM). Registered riders made 24,182 of those peak rentals, which is 97% of them.

Dashboard, anomaly detection and forecast

Interactive dashboard

The Dashboard sheet shows the whole story on one landscape screen, and it updates when any filter changes.



- **Filters:** drop-down lists for day type, weather, month and time band, plus From and To dates that work like a timeline. Slicers added to the final table in Excel also drive it.
- **KPI cards:** total, registered and casual rentals, average per hour, peak hour, busiest day, rain impact and the number of anomalies.
- **Visuals:** charts by hour, weekday, day and weather; an hour × weekday heatmap; a commute funnel; and eight key insights.

Anomaly detection

Each hour was compared with the normal demand for the same hour of the day, using a z-score (how many standard deviations it sits from that hour's average). Comparing like with like stops ordinary rush hours from being flagged.

With a threshold of 3, 7 of the 1,000 hours (0.7%) were flagged, all unusually busy. They are real busy hours rather than data errors, so they were kept and noted. The threshold sits in a yellow input cell on the Anomalies sheet, so it can be changed.

Forecast

Excel's FORECAST.ETS function (the engine behind Data › Forecast Sheet) predicts the next 48 hours. It learns the level, the trend and a 24-hour repeating pattern.

Tested on past hours, the average miss (MAE) is 21 rentals per hour. MASE is 0.98, which means the model is slightly better than simply repeating the last value. The forecast reliably shows *when* demand will be high or low, but not exact hourly numbers, because rain and holidays change demand suddenly.

Recommendations

1. **Guarantee bikes at commuter stations before 8 AM and 5 PM on weekdays.** These hours carry 43% of all rentals, and almost all of them are by registered riders.
2. **Rebalance and repair at night.** Between midnight and 5 AM demand averages 8 rentals per hour, so moving bikes then disturbs almost no one.
3. **Grow weekend and leisure use.** Casual riders are the weekend segment and favour warm afternoons. Weekend day passes, park and tourist routes, and first-ride offers for casual riders target them directly.
4. **Plan for rain.** Rainy hours lose about 46% of rentals. On rainy days reduce field staff, shift rebalancing, and promote "ride after the rain" offers once forecasts show clearing.
5. **Use the hourly forecast for staffing and bike placement** for the next day, focusing on the predicted peaks rather than exact counts.
6. **Keep loyal commuters.** Registered riders are 92% of rentals, so subscription retention and station reliability matter more than one-off promotions.

Limitations and next steps

The results describe six winter weeks, so they should be confirmed with more data before big decisions such as fleet size or seasonal pricing.

- Only 1,000 hours (1 January to 14 February 2011) are covered, all in season 1, so seasonal effects cannot be tested.
- 64 hours are missing from the timeline in the source data, and 14 February is a partial day.
- Temperature, humidity and wind speed are normalised with unknown units, so the results are relative.

- Heavy rain appears in a single hour, so no conclusion is possible for it.
- Correlation shows association, not cause.

Next steps would be to add a full year of data, include weather forecasts in the demand forecast, forecast daily totals (which are more stable than single hours), and track station-level demand.

Appendix: Excel functions and workbook guide

Purpose	Functions and features used	Where
Text	TEXT, LEFT, RIGHT, LEN, UPPER, & (join)	Final_Dataset (Record_ID, Month), Stats
Lookup	VLOOKUP, INDEX/MATCH (exact and approximate)	Dataset_A merge, weather and season labels, bands
Logic	IF, nested IF, AND, OR, IFERROR	Day type, working day, peak hour, demand level
Conditional	COUNTIF(S), SUMIF(S), AVERAGEIF(S)	Analysis, Dashboard_Calc, Stats
Statistics	AVERAGE, MEDIAN, MODE, STDEV, VAR, SKEW, KURT, QUARTILE, CORREL	Stats (descriptive statistics, correlation matrix)
Data quality	COUNTBLANK, SUMPRODUCT, ROWS, SUBTOTAL	Data_Quality, Final_Dataset (Visible)
Forecasting	FORECAST.ETS, FORECAST.ETS.CONFINT, FORECAST.ETS.STAT	Forecast
Presentation	Conditional formatting (colour scales, data bars), data validation drop-downs, Excel Table, charts, cell comments	All sheets
Automation	VBA module: one-click automated report, PDF export, filter reset, log	BikeSharingAutomation.bas

The workbook opens on a Cover sheet with a linked sheet index. The reading order is Dashboard, Insights, Analysis, Stats, Anomalies and Forecast, followed by the data stages: Final_Dataset, Dataset_A, Dataset_1 to 3, Data_Quality and Lookups. Cells with a small red

triangle carry a comment that explains the data, and blue numbers on yellow cells are inputs that can be changed.